

Auditing Local LLM Workflows for Older-Adult Referral Support

A High-Risk Healthcare AI Study on Evidence Attribution, Model Sensitivity,
Communication Stress Testing, and Data-Platform Readiness

High Risk Project
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Artifacts (Videos, Notebook, Report):

[GoogleDrive](#) | [GitHub](#)

AI 380H – AI in Healthcare
The University of Texas at Austin



Background & Motivation: Why Older-Adult Referral Support Matters

The real-world problem

Older adults often need referral decisions when the intake story is incomplete.

Common challenges:



Multiple chronic conditions

Memory, hearing, vision, or mobility barriers

Atypical or subtle symptoms



Caregiver-mediated history

Falls or symptoms that may be underreported

Why it matters: if the input story is fragile, an AI model may look confident while using incomplete context.

Research motivation in simple terms

1

Aging care is complex

Referral decisions depend on symptoms, history, vitals, and missing information.



2

LLMs are promising but risky

They can generate structured answers, but may hallucinate or over-trust weak narratives.



3

Our study audits before trust

We compare against TF-IDF, test Qwen/Phi local models, and stress-test degraded communication.

Why Japan and super-aging societies?

Japan context

29.3% of Japan's population was age 65+ in 2024

Super-aging societies need scalable, safe, data-aware healthcare AI.

[Source: Statistics Bureau of Japan](#)

Future Research Direction

Cloud + AI + Healthcare Data Platforms

This work connects technical infrastructure with safer AI workflows for aging-care needs, highlighting the importance of reliable data platforms, responsible model evaluation, and human-centered healthcare AI for super-aging societies.

Important boundary: the experiment uses MIMIC-IV-Ext-CDS, not Japanese clinical data.

Key message

This study is not building an autonomous referral doctor. It is an audit framework: test local LLMs, compare simple baselines, and examine failure modes before deployment claims.

Why Super-Aging Healthcare Needs Safer AI Workflows

Aging society creates a scale and safety problem

One patient story can be incomplete

Patient says

"I feel weak."



Caregiver adds

"He fell last week."



Clinician needs

Which specialist?

AI can help organize the case, but it can also amplify weak or incomplete information.

What responsible AI should do first



Compare with simple baselines

Do not assume LLMs beat TF-IDF.



Keep the input boundary explicit

Intake-stage support is not full-chart diagnosis.



Test communication degradation

Older-adult histories may be incomplete or mediated.



Separate auditability from correctness

Citing a section ID does not prove clinical support.

Societal contribution

For aging-care systems

- Reduce over-reliance on one subjective intake narrative
- Support safer review of referral suggestions
- Encourage longitudinal, structured health-data platforms
- Make AI evaluation more transparent before real-world use

Practical framing

This study treats healthcare AI as a workflow that must be audited, not a magic model that is trusted by default.

Key message

The social value is not only higher accuracy; it is safer evaluation, better data infrastructure, and clearer responsibility for AI-assisted aging care.

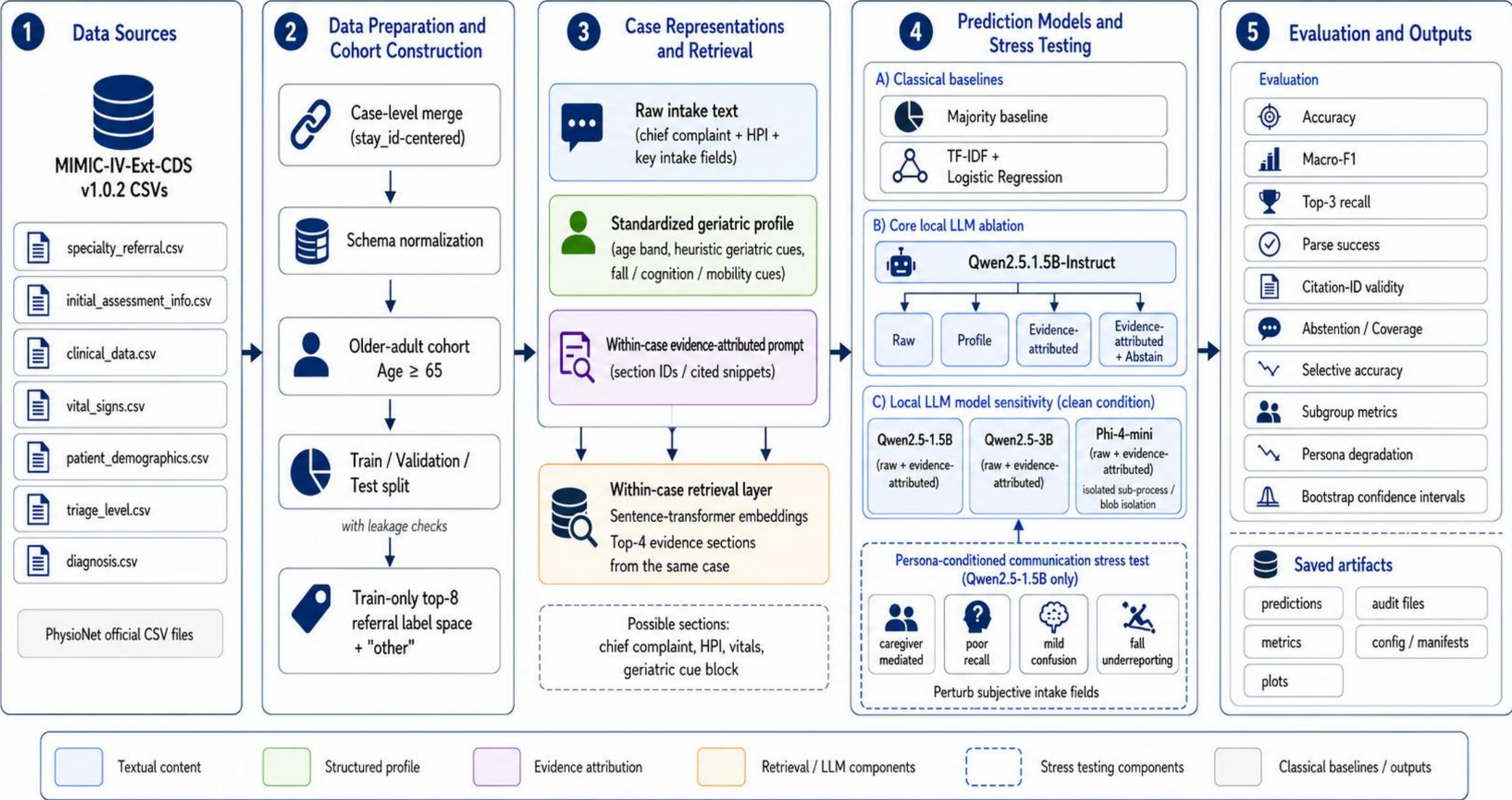
End to End Architecture

- Environment and Dataset
- Cohort and Referral Task
- Local LLM Strategy
 - Qwen2.5-1.5B
 - Qwen2.5-3B
 - Phi-4-mini



End-to-End Experimental Architecture

From MIMIC-IV-Ext-CDS cases to referral labels, local LLM workflows, and robustness audits



Environment and Dataset: Reproducible Local Evaluation

Execution environment

1

Google Colab Pro

Notebook execution environment

2

T4 GPU

Practical local LLM inference, faster than CPU

3

Google Drive

Persistent storage for predictions, metrics, plots, audit files

4

Local inference

Avoids sending MIMIC-derived text to external LLM APIs

MIMIC-IV-Ext-CDS benchmark

Why this dataset?

- MIMIC-derived clinical decision-support benchmark
- Designed for triage, specialist referral, and diagnosis tasks
- Provides structured CSV files that can be merged into case-level examples
- Useful for reproducible audit-style evaluation

Dataset scale in paper

9,150 real-world medical cases

2,200 refined task-specific cases

Artifacts saved for reproducibility

processed datasets

metrics

case profiles

plots

retrieval cache

audit files

predictions

config manifests

Why this matters

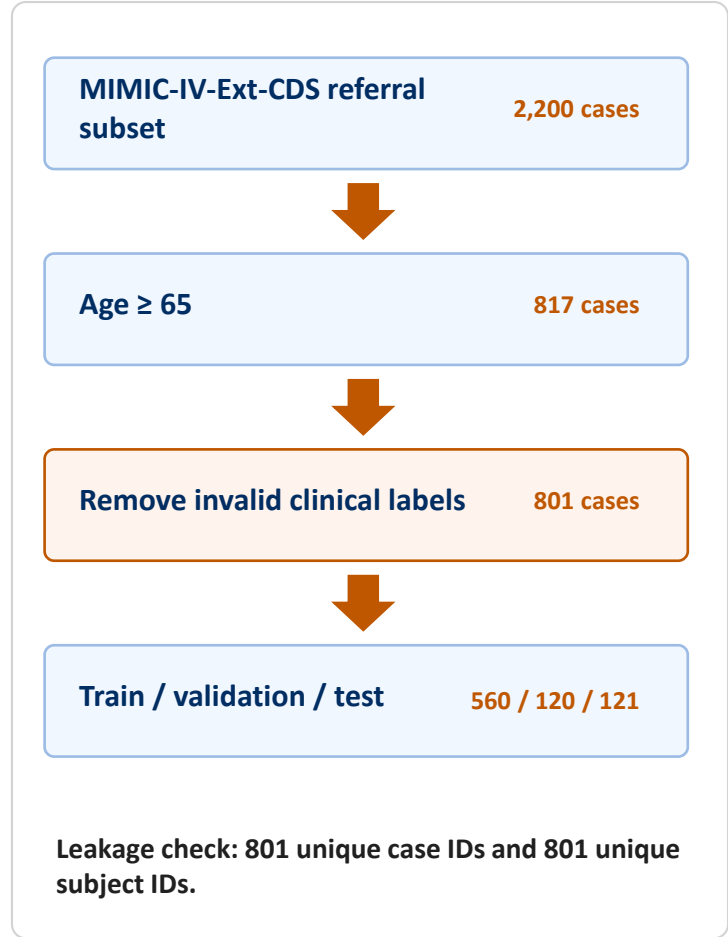
The same cases, prompts, predictions, and metrics can be checked again later. This is important in high-risk AI because results must be explainable and traceable.

Key message

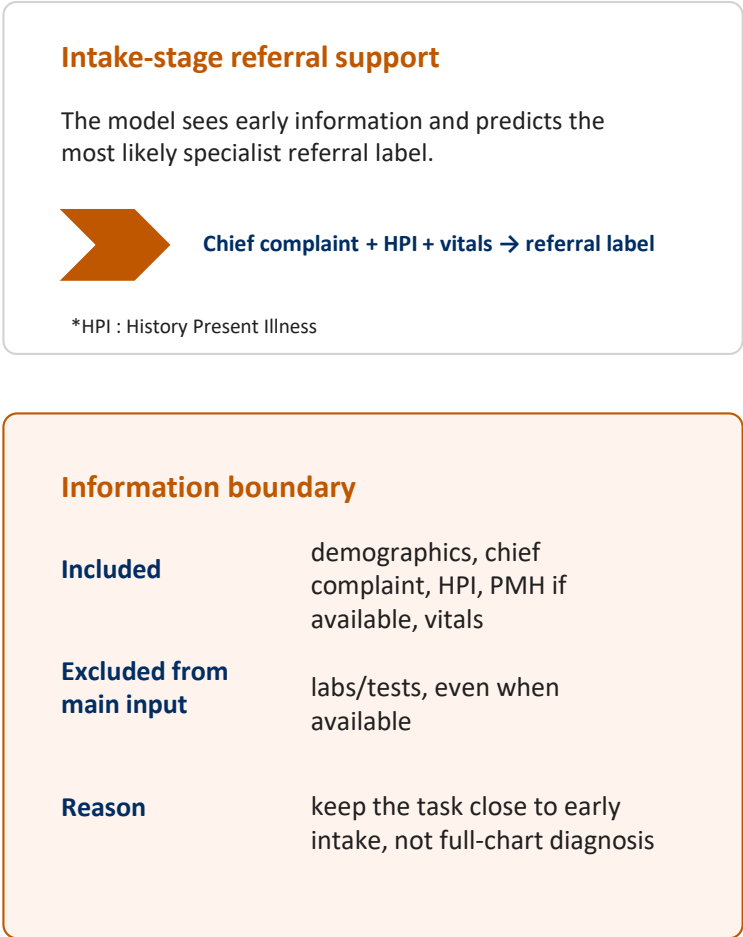
The environment was designed for local, reproducible experimentation: Colab Pro + T4 GPU for inference, and Google Drive for persistent artifacts.

Cohort and Referral Task: From Dataset to Older-Adult Cases

Cohort construction



Target decision point



What the cohort looks like



Referral labels are imbalanced; chief complaint and HPI are complete, while PMH/medication fields are unavailable in this extract.

Key message The study focuses on a clearly defined older-adult intake-stage task: predict referral specialty from early patient information, not from the full chart.

Local LLM Strategy: Qwen, Phi, Prompting, and Stress Testing

Why these local LLMs?

Qwen2.5-1.5B Core local model

Full prompt-arm comparison and persona stress test.

Qwen2.5-3B Same-family scaling

Tests whether larger Qwen capacity changes the conclusion.

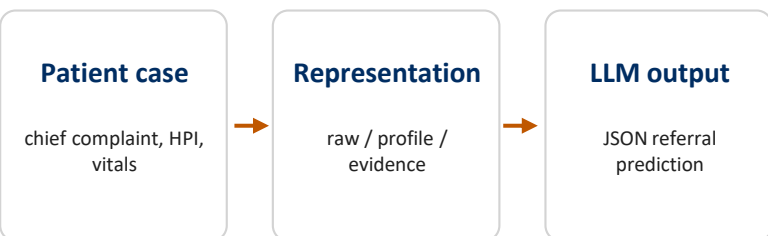
Phi-4-mini Cross-family sensitivity

Tests whether the pattern is specific to Qwen. Run with isolated process for safety.

Practical reason: small-to-moderate instruction models that can run locally in Colab-style environments.



Prompting approach



Three ways to show the same case

	Raw intake text	direct chief complaint + HPI
	Geriatric profile	structured sections + fall/confusion/mobility cues
	Evidence-attributed prompt	retrieves same-case sections and asks for cited section IDs

Not external RAG: retrieval is from the same patient case only.

Stress testing and evaluation

Communication perturbations

poor recall

mild confusion

caregiver-mediated history

fall underreporting

Evaluation signals

- accuracy + macro-F1
- top-3 recall
- parse success
- citation-ID validity
- abstention / coverage
- persona degradation

Most important mindset

We are evaluating the workflow, not only the model name.

Key message The LLM experiment tests whether local models and prompting workflows remain useful, auditable, and robust when older-adult communication becomes imperfect.

Prediction Models: Why Baselines, Qwen, and Phi Matter

This part explains what each model group is, why it is included, and how it helps interpret the experiment.

A) Classical baselines

What it is:
Simple reference models used to check whether a more advanced model is actually adding value.

Majority baseline = minimum benchmark
TF-IDF + Logistic Regression = strong transparent text baseline

Why it matters: prevents overclaiming LLM performance.

*TF-IDF = Term Frequency–Inverse Document Frequency

B) Local LLM families

Qwen2.5-1.5B Core local model

Full prompt-arm comparison and persona stress test.

Qwen2.5-3B Same-family scaling

Tests whether larger Qwen capacity changes the conclusion.

Phi-4-mini Cross-family sensitivity

Tests whether the pattern is specific to Qwen. Run with isolated process for safety.

Qwen tests the main workflow and scaling; Phi checks whether results generalize across model families.

How the Qwen 1.5B core ablation works

The main experiment compares several ways of presenting the same patient case to the model.

Raw

Plain intake text prompt.

Basic LLM reference.

Profile

Structured geriatric profile.

Tests whether clearer organization helps.

Evidence-attributed

Within-case evidence sections and citation IDs.

Separates auditability from accuracy.

Evidence + Abstain

Model may abstain when evidence is insufficient.

Tests selective prediction behavior.

Key message

We compare local LLMs against strong simple baselines, then ask whether prompt design, model size, and model family change the conclusion.

Results



Results Overview: What Did We Learn?

Main result

The simple TF-IDF Logistic Regression baseline was the strongest clean classifier; local LLMs added structured outputs and audit features, but did not outperform the baseline.

1 Core performance

Which model predicted referrals best?

- TF-IDF LR achieved the best clean accuracy and Macro-F1.

2 Model sensitivity

Did larger / different local LLMs change the conclusion?

- Qwen2.5-3B and Phi-4-mini improved but did not beat TF-IDF.

3 Audit behavior

Did evidence citation and abstention help?

- Citation traceability improved, but clean prediction did not.
- Prompt-only abstention was not reliable for selective prediction.

4 Stress testing

Which older-adult communication scenario was most fragile?

- Caregiver-mediated histories caused the largest degradation.

Key message

The Results section is not only a score report; it is an audit of performance, traceability, and robustness.

Before the Scores: What Was Evaluated?

Cohort and task

Cohort

801

older-adult cases

Test

121

held-out cases

Age

≥65

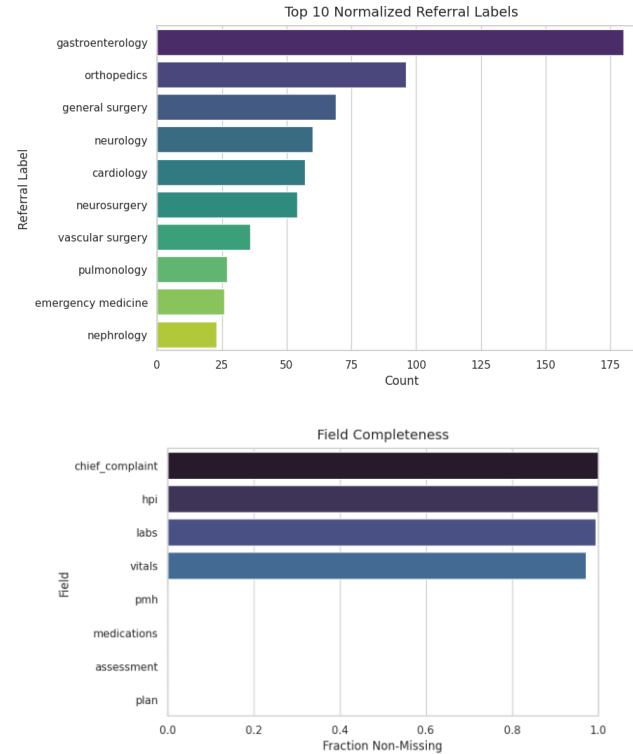
years

- Task: predict specialist referral label.
- Input: intake-stage information only.
- Labs/tests existed but were excluded from the main representation.

Why this matters

The models are tested early in the care pathway, not after a full clinical workup.

Data context



Referral labels were imbalanced. Chief complaint and HPI were complete, vitals were mostly available, while PMH/medications were unavailable in the extracted representation.

How to read the metrics

Accuracy

correct top answer

Macro-F1

balanced across labels

Top-3 recall

correct label appears in top 3

Citation-ID validity

cited section exists in prompt

**Primary interpretation: Accuracy + Macro-F1.
Audit interpretation: Top-3 + citation validity.**

Key message

Because referral labels are imbalanced, Macro-F1 is essential; because LLMs generate structured outputs, audit metrics are also needed.

Core Clean-Condition Result: TF-IDF Was Strongest

Main performance table

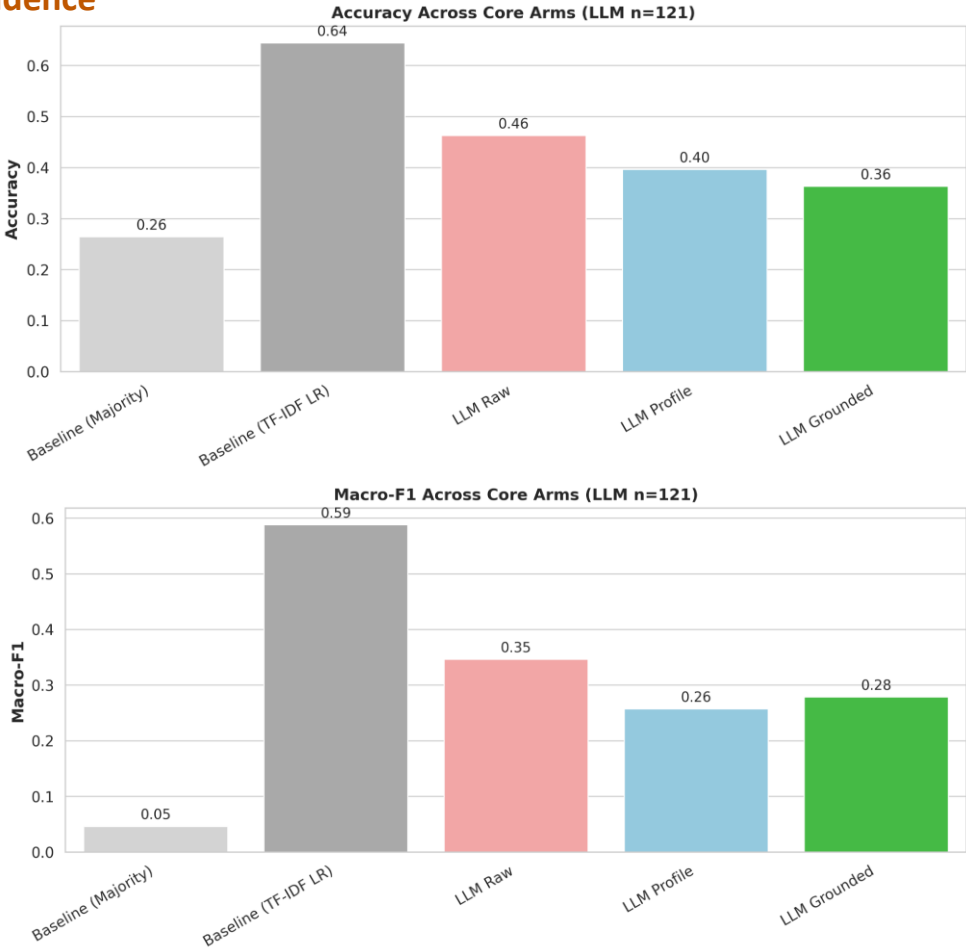
Model	Acc.	Macro-F1	Top-3
Majority baseline	0.264	0.046	—
TF-IDF LR	0.645	0.588	0.884
Qwen 1.5B Raw	0.463	0.346	0.669*
Qwen 1.5B Profile	0.397	0.257	0.595*
Qwen 1.5B Evidence	0.364	0.279	0.612*
Qwen 1.5B Abstain	0.298	0.151	0.570*

*LLM top-3 recall is post-processed and exploratory.

Simple reading

- TF-IDF LR was the strongest clean classifier.
- The best Qwen2.5-1.5B arm was raw prompting.

Visual evidence



Both accuracy and Macro-F1 show the same pattern:
the simple lexical baseline outperformed the local Qwen2.5-1.5B prompt arms.

Key message

The LLM produced structured answers, but the simple TF-IDF model predicted referral labels more accurately in the clean setting.

Model Sensitivity: Bigger and Different LLMs Helped, But Did Not Win

Question tested

If we use Qwen2.5-3B or Phi-4-mini, do the clean-condition findings change?

TF-IDF LR

Reference baseline

Accuracy	0.645
Macro-F1	0.588
Top-3	0.884

Qwen2.5-1.5B Raw

Core local LLM

Accuracy	0.463
Macro-F1	0.346
Top-3	0.669*

Qwen2.5-3B Raw

Best local LLM

Accuracy	0.537
Macro-F1	0.447
Top-3	0.818*

Phi-4-mini Raw

Cross-family LLM

Accuracy	0.496
Macro-F1	0.432
Top-3	0.719*

What improved?

Qwen2.5-3B and Phi-4-mini were stronger than Qwen2.5-1.5B raw.

What stayed the same?

Neither local LLM exceeded the TF-IDF Logistic Regression baseline.

Why it matters

The conclusion is not based only on one small local model.

Key message

Scaling and cross-family testing strengthened the conclusion: local LLMs improved, but the lexical baseline remained strongest.

Audit Behavior: Evidence Attribution Helped Traceability, Not Prediction

Evidence-attributed prompting

What it does

The model receives retrieved sections from the same case and must cite section IDs.

Model	Raw F1	Evidence F1	Citation-ID
Qwen2.5-1.5B	0.346	0.279	1.000
Qwen2.5-3B	0.447	0.368	1.000
Phi-4-mini	0.432	0.400	0.983

Interpretation

Citation traceability was high, but raw prompting had better Macro-F1 for clean classification.

Abstention behavior

What abstention tries to do

The model can avoid a forced prediction when evidence seems insufficient.

Coverage

0.893

answered cases

Selective accuracy

0.324

after abstention

Interpretation

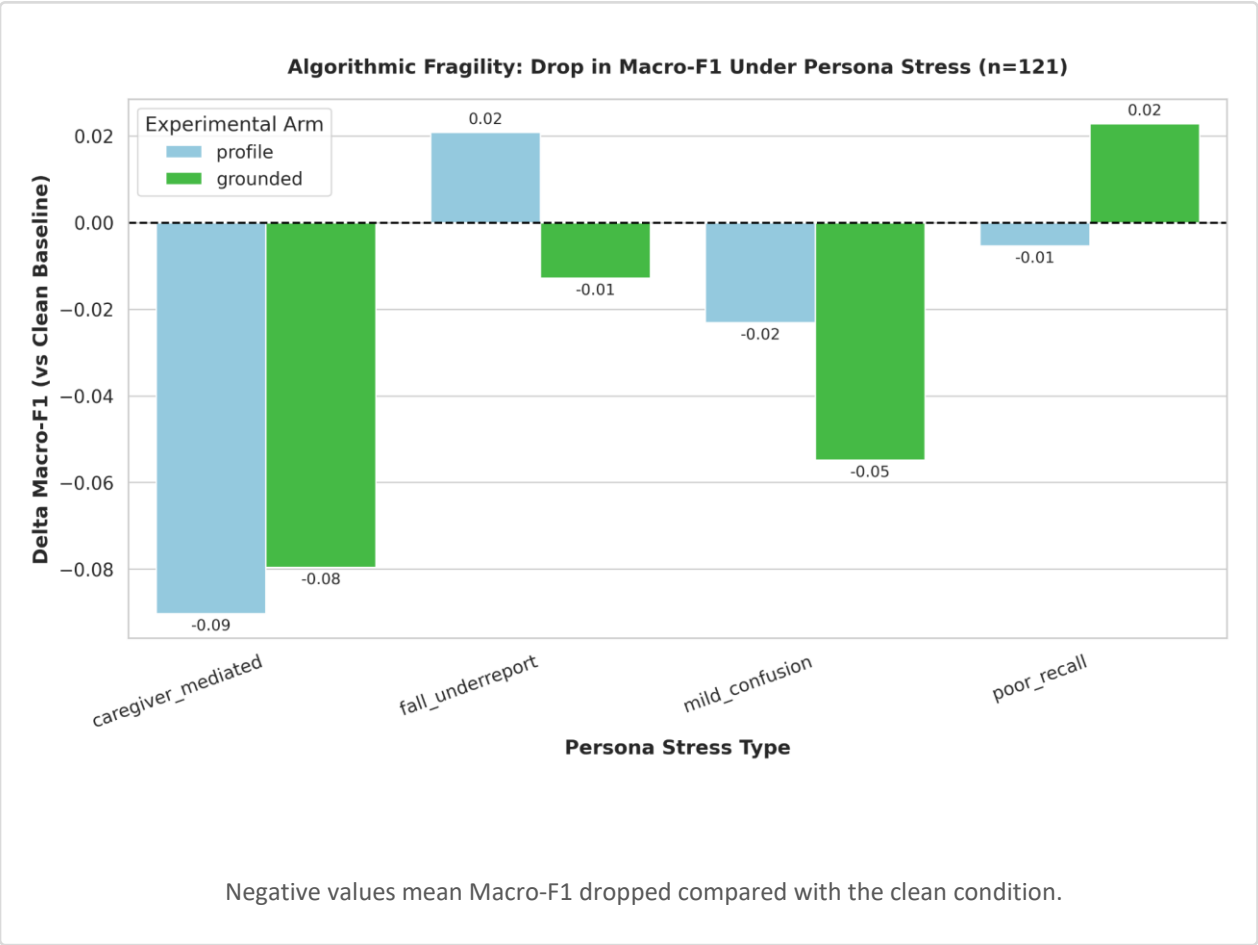
A prompt instruction alone did not create a reliable uncertainty filter.

Key message

Evidence attribution improved audit structure; abstention and citation should not be treated as proof of clinical correctness.

Communication Stress Test: Caregiver-Mediated History Was Most Fragile

The stress test asks whether performance changes when older-adult communication is degraded.



Most important stress result

Largest degradation

Caregiver-mediated history caused the biggest Macro-F1 drop in both tested arms.

Arm	Clean F1	Stress F1	Drop
Profile	0.257	0.167	-0.090
Evidence	0.279	0.199	-0.080

Why this matters

Older-adult histories may be indirect, incomplete, or filtered through caregivers. The model became fragile when the communication channel changed.

Key message

The stress-test result shows why clean text performance is not enough for high-risk older-adult referral support.

Results Takeaway: What the Experiment Supports

A concise interpretation for the end of the Results section.

1 Do not skip simple baselines

TF-IDF Logistic Regression was strongest in clean classification.

2 Local LLMs improved with scaling

Qwen2.5-3B and Phi-4-mini improved over Qwen2.5-1.5B but still trailed TF-IDF.

3 Traceability is not accuracy

Evidence attribution improved citation structure, but did not improve clean prediction.

4 Prompt-only abstention is weak

The abstention arm did not create a reliable selective-prediction mechanism.

5 Communication stress matters

Caregiver-mediated history was the most consistently harmful perturbation.

Final Result

Local LLM workflows for older-adult referral support should be benchmarked against strong simple baselines, tested across model families, and stress-tested under communication degradation before deployment claims are made.

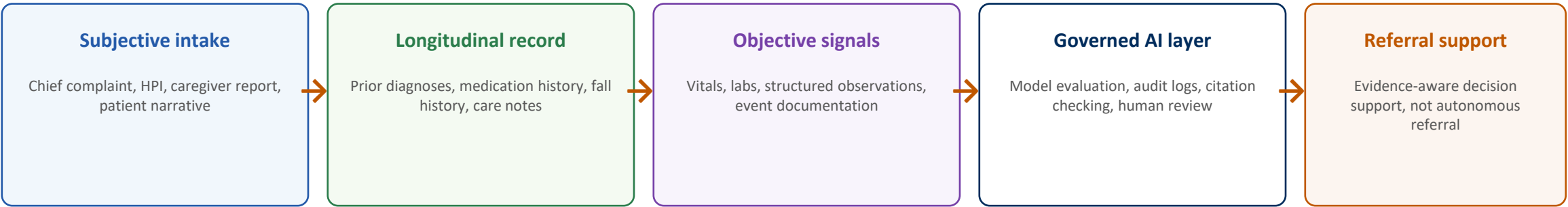
Implications for Health Data Platforms

The caregiver-mediated failure mode suggests that future AI support should not depend only on one subjective intake story.

Observed failure mode

Caregiver-mediated histories produced the largest degradation. This implies that a referral-support workflow may become fragile when it relies too heavily on one incomplete or indirect narrative.

Future platform direction: build a more stable information substrate



What the experiment supports

- Single-channel intake narratives can be fragile.
- AI workflows should be tested under realistic communication degradation.

What the experiment does not prove

- It does not prove a health-data platform will solve referral fragility.
- It motivates future evaluation of integrated, longitudinal, interoperable data platforms.

Key message

Future healthcare AI platforms should combine narrative, structured, temporal, and clinician-reviewed context before AI support is trusted in aging-care workflows.

Conclusion



Limitations and Future Work

These limitations keep the conclusion responsible and define the next research steps.

Main limitations

Dataset

MIMIC-IV-Ext-CDS is not Japanese clinical data; referral labels and top-eight + other label collapse simplify real referral practice.

Models

Models were small or moderate local models selected for Colab feasibility; Qwen3B and Phi were evaluated only in raw and evidence-attributed arms.

Evaluation

Top-3 recall is exploratory; citation-ID validity is not clinical evidence support; clinician-adjudicated evidence review was not completed.

Stress test

Persona perturbations are deterministic English transformations, not validated patient or caregiver simulations; evidence-attributed stress used clean retrieval sections.

Future work

1

Model expansion

Test larger local, medically tuned, and calibrated models.

2

Evidence validation

Ask clinicians to judge whether cited evidence actually supports the referral decision.

3

Retrieval robustness

Compare clean-evidence buffering, objective-only retrieval, and fully persona-perturbed retrieval.

4

Clinical realism

Evaluate clinician-approved referral subsets and compare intake-stage versus full-chart settings.

5

Platform research

Study whether longitudinal data platforms reduce dependence on fragile subjective narratives.

Interpretation: this is a retrospective audit framework, not a clinical deployment claim.

Conclusion and Final Key Takeaway

The study provides a responsible evaluation framework for local LLM referral-support workflows in older-adult cases.

Final Conclusion

Local LLM workflows were useful for audit-oriented evaluation, but in this benchmark they did not outperform a strong TF-IDF logistic-regression baseline for clean older-adult referral classification.

1 Baseline first

Simple TF-IDF LR remained the strongest clean-condition model.

2 Scaling helped

Qwen2.5-3B and Phi-4-mini improved LLM results, but did not exceed TF-IDF.

3 Audit ≠ accuracy

Evidence attribution improved citation traceability, not prediction.

4 Abstention needs calibration

Prompt-only abstention did not produce reliable selective prediction.

5 Stress testing matters

Caregiver-mediated history was the most harmful perturbation.

Final message

For older-adult referral support, deployment claims should come only after strong baseline comparison, explicit input-boundary control, model-family sensitivity analysis, communication stress testing, and separate review of evidence citation versus clinical correctness.

Longer-term direction:

Future work should investigate whether integrated health-data platforms can provide a more stable information substrate for AI-assisted aging-care workflows by combining subjective intake text with longitudinal structured data, prior diagnoses, medication history, fall documentation, objective measurements, and clinician-reviewed context.

Thank you!

